

LONDON WC PENDEREL HOUSE, United Kingdom

WMO: 037780

Lat: 51.517N

Lon: 0.117W

Elev: 23

StdP: 101.05

Time Zone: 0.00 (EUW)

Period: 82-92

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-2.6	-1.0	-6.7	2.2	-0.5	-5.3	2.4	0.8	14.0	9.5	13.0	6.4	5.8	40	0.605

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.8	27.2	18.9	25.6	18.0	24.1	17.4	19.9	25.8	18.9	23.9	18.1	22.6	4.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.5	12.6	22.3	16.8	12.0	21.6	16.1	11.5	20.6	56.8	25.9	53.5	24.0	50.9	22.4	24.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.7	10.3	9.4	-3.2	30.0	3.1	2.5	-5.5	31.9	-7.3	33.3	-9.0	34.8	-11.3	36.6		
(5)			-4.5	21.7	2.8	1.5	-6.5	22.8	-8.2	23.7	-9.8	24.5	-11.8	25.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.7	5.8	5.4	8.0	9.7	13.0	15.8	18.8	18.5	15.8	12.8	8.7	7.4
	DBStd	5.43	3.40	3.73	2.76	2.95	2.97	2.91	2.68	2.68	2.49	2.45	3.00	2.81
	HDD10.0	535	131	131	73	41	6	0	0	0	0	4	61	88
	HDD18.3	2526	387	361	320	261	168	88	27	33	81	173	289	338
	CDD10.0	1152	2	3	12	31	99	174	273	262	176	91	21	9
	CDD18.3	101	0	0	0	0	2	13	42	37	7	1	0	0
	CDH23.3	495	0	0	0	0	13	59	239	161	21	1	0	0
	CDH26.7	78	0	0	0	0	1	6	46	26	1	0	0	0
Wind	WSAvg	5.0	5.7	5.7	5.5	5.2	4.8	4.7	4.5	4.3	4.4	4.9	5.0	5.3
Precipitation	PrecAvg	586	51	34	46	44	48	47	44	49	55	60	56	53
	PrecMax	801	126	89	98	106	88	107	92	118	146	137	139	113
	PrecMin	437	9	7	4	9	1	6	9	7	4	3	17	12
	PrecStd	86	27	19	22	24	25	30	24	28	39	42	31	25
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.1	14.3	17.8	21.1	25.1	27.5	30.6	29.6	25.7	21.4	16.1	14.2
		MCWB	10.9	10.8	12.1	13.5	17.0	19.7	20.1	20.2	17.7	16.3	13.3	11.8
	2%	DB	12.0	12.6	14.7	18.7	22.4	25.0	27.7	26.6	23.5	18.8	15.0	13.4
		MCWB	10.2	9.9	10.5	12.3	15.5	17.4	19.1	18.6	17.2	15.3	13.0	11.4
	5%	DB	11.3	11.5	13.2	16.5	20.3	23.1	26.0	25.0	21.7	17.6	14.1	12.7
		MCWB	9.5	9.1	9.7	11.3	14.2	16.8	18.2	17.8	16.1	14.3	12.3	10.8
	10%	DB	10.5	10.5	11.9	14.7	18.3	21.1	24.2	23.5	20.0	16.4	13.1	11.7
		MCWB	8.7	8.3	8.8	10.2	13.1	15.8	17.5	17.3	15.4	13.6	11.3	10.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.5	11.6	12.3	14.7	17.7	20.4	21.2	20.9	19.2	16.8	14.2	12.6
		MCDB	12.7	13.4	16.5	19.3	23.6	25.9	28.5	28.5	23.7	19.9	15.6	13.6
	2%	WB	10.6	10.5	11.2	12.9	16.0	18.8	20.0	19.6	17.7	15.8	13.4	11.8
		MCDB	11.8	12.2	14.1	17.7	21.8	23.5	26.5	25.1	21.5	18.5	14.8	13.0
	5%	WB	9.8	9.5	10.3	11.8	14.7	17.5	19.0	18.6	17.0	14.9	12.5	11.1
		MCDB	11.1	11.2	12.6	15.7	19.8	21.9	24.4	23.3	20.4	16.8	13.9	12.5
	10%	WB	8.9	8.6	9.5	10.7	13.6	16.2	18.0	17.9	16.2	14.1	11.5	10.2
		MCDB	10.4	10.3	11.5	14.3	17.7	20.5	22.9	22.2	19.5	16.0	13.0	11.7
Mean Daily Temperature Range	5% DB	MDBR	4.1	4.3	5.5	6.8	7.1	7.0	7.8	7.4	6.5	5.3	4.3	3.8
		MCDBR	4.7	5.2	7.0	9.4	10.4	9.9	10.8	9.8	9.0	6.1	4.5	4.1
		MCWBR	4.6	4.0	4.2	4.9	5.2	5.1	4.8	4.3	4.2	3.5	3.6	3.8
	5% WB	MCDBR	4.8	5.1	6.2	8.6	9.8	9.4	10.0	8.8	7.7	5.8	4.5	4.2
		MCWBR	5.0	4.5	4.2	4.9	5.4	5.1	5.0	4.2	4.4	4.2	3.9	4.1
Clear-Sky Solar Irradiance	taub	0.331	0.347	0.384	0.405	0.410	0.411	0.407	0.397	0.379	0.362	0.335	0.321	
	taud	2.369	2.373	2.264	2.225	2.251	2.276	2.294	2.340	2.389	2.419	2.414	2.355	
	Ebn at Noon	672	770	802	830	844	845	842	831	804	742	664	618	
	Edh at Noon	63	82	109	126	129	127	123	112	95	77	60	55	
All-Sky Solar Radiation	RadAvg	0.80	1.46	2.56	4.00	4.81	5.33	5.05	4.20	3.17	1.83	0.96	0.63	
	RadStd	0.07	0.20	0.34	0.43	0.32	0.49	0.48	0.34	0.26	0.17	0.11	0.08	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (81 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page